

```
import nltk
nltk.download('wordnet')
nltk.download('omw-1.4')
```

```
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
True
```

```
from nltk.stem import PorterStemmer

ps = PorterStemmer()

words = ["playing", "played", "plays", "studies", "studying",
         "connected", "connection", "computers"]

print(f"{'Original Word':<15}{'Stemmed Word'}")
print("-" * 30)
for w in words:
    print(f"{w:<15}{ps.stem(w)}")
```

Original Word	Stemmed Word
playing	play
played	play
plays	play
studies	studi
studying	studi
connected	connect
connection	connect
computers	comput

```
from nltk.stem import WordNetLemmatizer

lemmatizer = WordNetLemmatizer()

words = ["cats", "dogs", "running", "runs", "ran", "studies",
         "studying", "better", "children", "mice", "went", "ate",
         "leaves", "caring"]

print(f"{'Original Word':<15}{'Lemma (noun)':<15}{'Lemma (verb)'}")
print("-" * 45)
for w in words:
    lemma_n = lemmatizer.lemmatize(w, pos='n')
```

```
lemma_v = lemmatizer.lemmatize(w, pos='v')
print(f"{w:<15}{lemma_n:<15}{lemma_v}")
```

Original Word	Lemma (noun)	Lemma (verb)

cats	cat	cat
dogs	dog	dog
running	running	run
runs	run	run
ran	ran	run
studies	study	study
studying	studying	study
better	better	better
children	child	children
mice	mouse	mice
went	went	go
ate	ate	eat
leaves	leaf	leave
caring	caring	care

```
import re

text = """Welcome to the Natural Language Processing workshop! For more informati
https://www.nlpworkshop.com or www.python.org. You can contact the coordinator at
nlpworkshop@gmail.com or support@python.org. For registration, call +91-987654321
9123456789. Follow us on social media @NLPWorkshop and @PythonLearner. Share your
experience using #NLP, #Python, and #MachineLearning. You can also visit
https://github.com/NLPWorkshop for the latest updates."""

print("Input Text:\n", text)
print("\n" + "="*60 + "\n")

emails = re.findall(r'[\w.+-]+@[\w-]+\.[a-zA-Z]{2,}', text)
print("1. Email Addresses:", emails)

urls = re.findall(r'https?://[^\s.]+(?:\.[^\s.]+)*|www\.[^\s.]+(?:\.[^\s.]+)*'
print("2. URLs:", urls)

mobiles = re.findall(r'(?:\+91[-\s]?)?\d{10}', text)
print("3. Mobile Numbers:", mobiles)

hashtags = re.findall(r'#\w+', text)
print("4. Hashtags:", hashtags)

mentions = re.findall(r'(?<![\w.])@\w+', text)
print("5. Mentions:", mentions)
```

Input Text:

Welcome to the Natural Language Processing workshop! For more information, visit <https://www.nlpworkshop.com> or www.python.org. You can contact the coordinator at nlpworkshop@gmail.com or support@python.org. For registration, call +91-9876543210 or 9123456789. Follow us on social media @NLPWorkshop and @PythonLearner. Share your experience using #NLP, #Python, and #MachineLearning. You can also visit <https://github.com/NLPWorkshop> for the latest updates.

=====

1. Email Addresses: ['nlpworkshop@gmail.com', 'support@python.org']
2. URLs: ['<https://www.nlpworkshop.com>', 'www.python.org', '<https://github.com/NLPWorkshop>']
3. Mobile Numbers: ['+91-9876543210', '9123456789']
4. Hashtags: ['#NLP', '#Python', '#MachineLearning']
5. Mentions: ['@NLPWorkshop', '@PythonLearner']